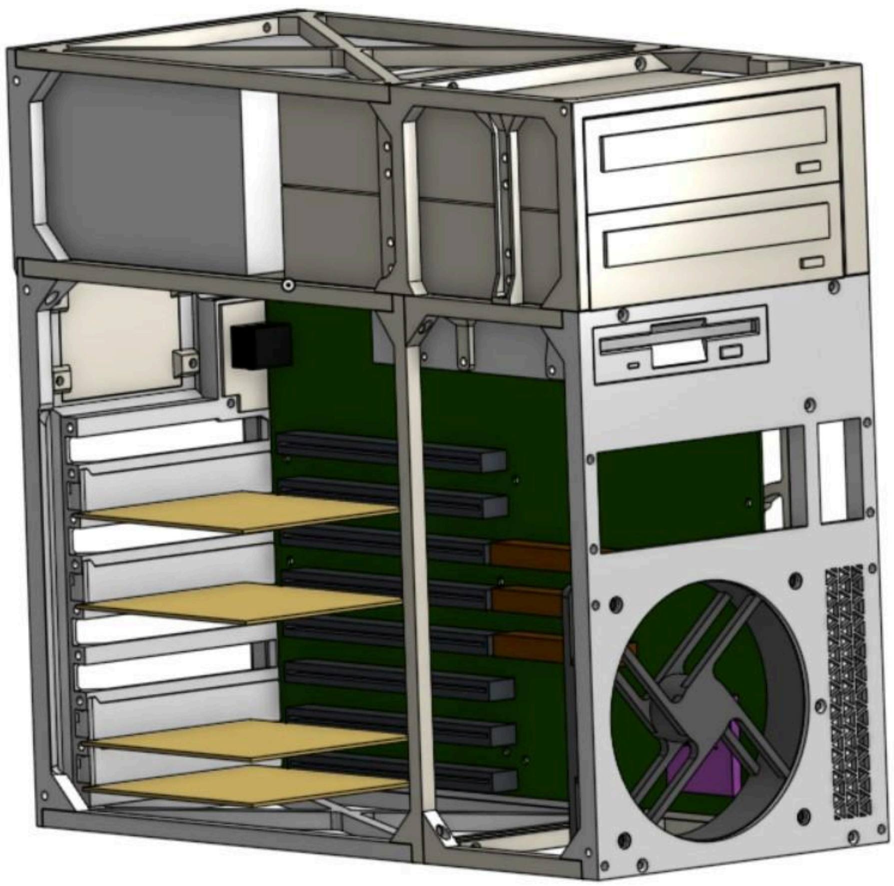
ModularAT

User's guide



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A. Introduction

This is a fully 3D printable computer case, which is designed to fit motherboards in the Baby AT form factor. One does not need to salvage other pre-defined sub assemblies from original cases, such as drive cages or motherboard trays are not required in the build process. All the essential structure parts are included in the repository.

1. *What does this fit?*

In a short answer, pretty much *any Baby AT motherboard* out there. Yes, even ones that have more than just a full size DIN plug for the keyboard, or which only have PS/2 ports. The case has a large opening at the back with a swappable plate, which allows for a flexible configuration, including a blank plate if the build is around an SBC + backplane. Most options out there are covered. Length wise, it can fit the maximum Baby AT size (up to 330mm) when using the frame extensions

An important factor to keep in mind is that the Baby AT form factor is not very well defined across the decade or so it was used throughout the late 80s and the first half of the 90s. Many motherboards from the 386 era (and older) have a tendency to not respect some of the hole positions when compared to most of the newer boards which followed them. As such, making a “one glove fits all hands” motherboard tray is very difficult to achieve without significantly increasing the build complexity. I have done measurements on my own board collection and discovered that the mounting holes can differ in position by as little as one millimeter!

Knowing all of this, I have decided the best approach is to stick to the most commonly used hole pattern, described in the *Upgrading and Repairing PCs 10th Edition* by Scott Mueller (great book by the way, I recommend giving it a glance over if you haven’t done so yet). It is inevitable some early boards will not match all the holes, but the motherboard spacers can be removed in case component legs exist in those spots.

2. *Parts compatibility*

The case can accommodate up to *eight full height, full length expansion cards* in its big configuration (they are screwed down using normal M3 PC screws). It can fit any *standard ATX power supply* (unusually long ATX units also fit, but they will eat space from the drive bay). *Two 5.25 inch drive bays* are provided, with support for long CD-ROMs or the typical 360K/1.2M floppy drives. Additionally, *one 3.5 inch drive bay* is provided underneath, for use with a typical 1.44M floppy drive (or any 3.5 inch bay drive accessory that uses floppy drive mounting holes). Up to *three 3.5 inch hard drives* can be mounted behind the motherboard tray. The reason for that number is simple: the 4th spot is reserved for PSU cable management, those cables have to go somewhere as well and it would be a shame to have them fill up the motherboard area, like most of the cases from that era.

3. *Brass inserts*

Initially, I designed the frame around conventional M3 hexagonal nuts that you can find in most hardware stores. This turned out to not be a great choice however, as it made the frame too thin in many places and prone to fracturing. In the next version, I used brass inserts which turned out to be way easier and stronger, without having to make the frame needlessly thick.

For ease of installation, a jig tool for brass inserts is recommended (though not required). The bare minimum you will need is an old soldering iron, for which you won't care if the tip gets burnt up with plastic. In that case, I also recommend shaving the iron tip with a file or sandpaper so it can fit better inside the brass insert and make better contact with its surface.

A good technique I've seen online is to slowly melt it down in the part until you can just barely see part of the insert still exposed, but not fully seated in. Then you quickly come with a flat metallic object: knife, bigger bolt with a flattened head, or really, anything with a nice and flat surface which can press the insert the rest of the way down and make it flush with the surface of the part it gets embedded in. The end result will almost always be a perfectly flush brass insert.

B. Bill of materials

You will need the following to build the frame:

- one kg of PLA/PETG/ABS spool (or any material of choice)
- ?x M3 x 4.2mm (outer diameter) x 4mm (depth) brass inserts
- ?x M3 x 4mm countersunk screws
- ?x M3 x 8mm countersunk screws
- ?x M3 x 12mm countersunk screws
- ?x M3 x 16mm countersunk screws

It’s also good to have mounting hardware for the components:

- 4x 7/32” self tapping computer fan screws
- a handful of M3 motherboard screws
- a handful of #6-32 UNC (hard drive) screws

C. Getting started

Have all the 3D models in the frame subdirectory printed and ready to use. From the accessories subdirectory you will need at least a few motherboard spacers.

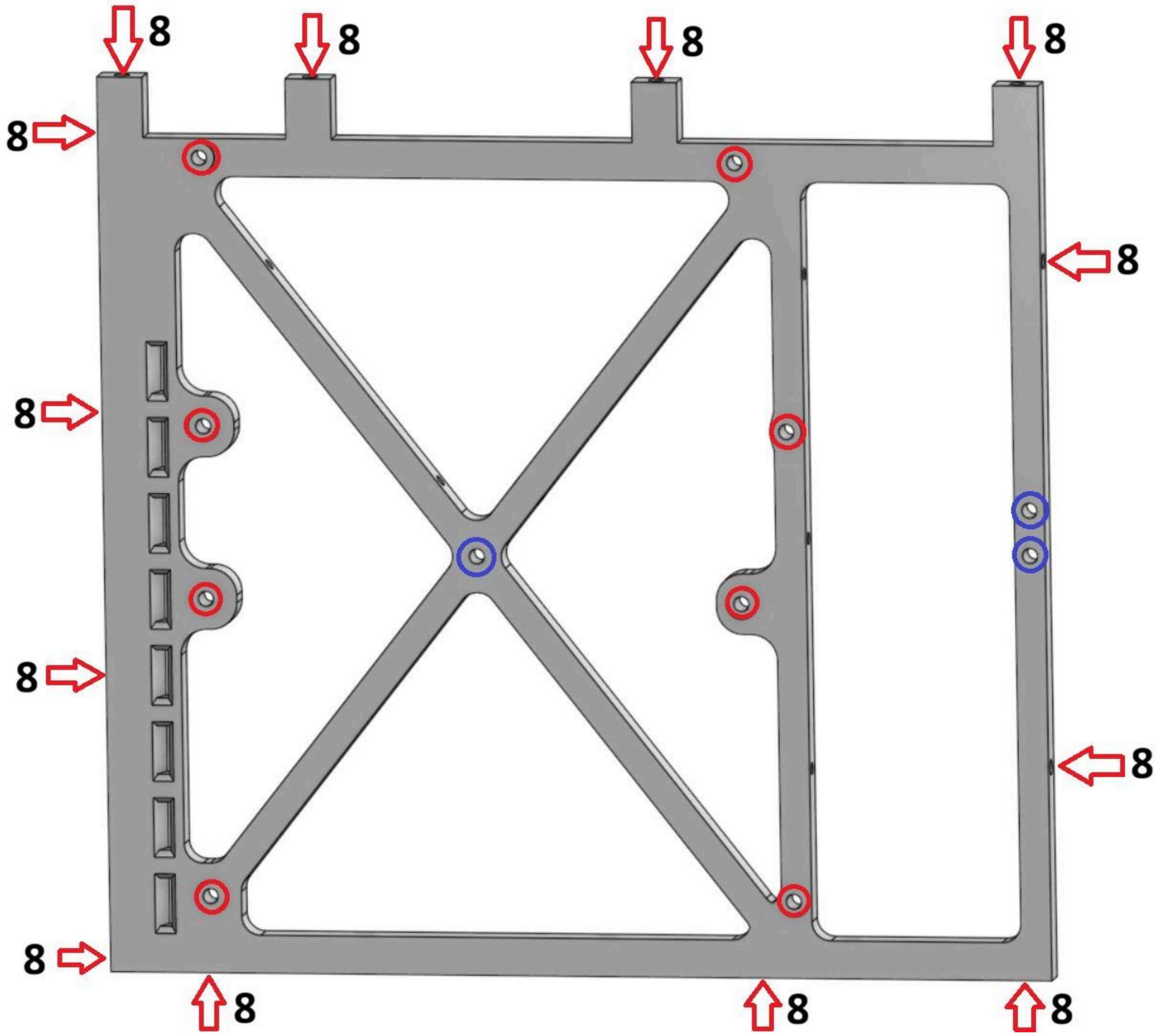
For all the brass insert diagrams, arrows and/or circles will point to their destination. Arrows represent the direction from which the brass insert should be installed, while circles indicate the brass inserts should be inserted from the POV of the diagram (as seen). Pay attention to the colors of the markers and any remarks underneath the diagram, there can be special cases (mount from the other side, etc).

If a number is provided next to the marker, it represents the <u>*max depth of the hole*</u> in millimeters. This will be useful both for picking brass inserts with the right depth, as well as picking the screws with the right length for final assembly.

If a number is **NOT** provided next to the marker, it means the brass inserts and/or screws can be of <u>*any length*</u>, as long as they don’t run into other components you may install later (for e.g. the motherboard holes are see thru, but if there is a hard drive mounted on the backside, it could be damaged by screws which are too long in that area).

1. Base frame

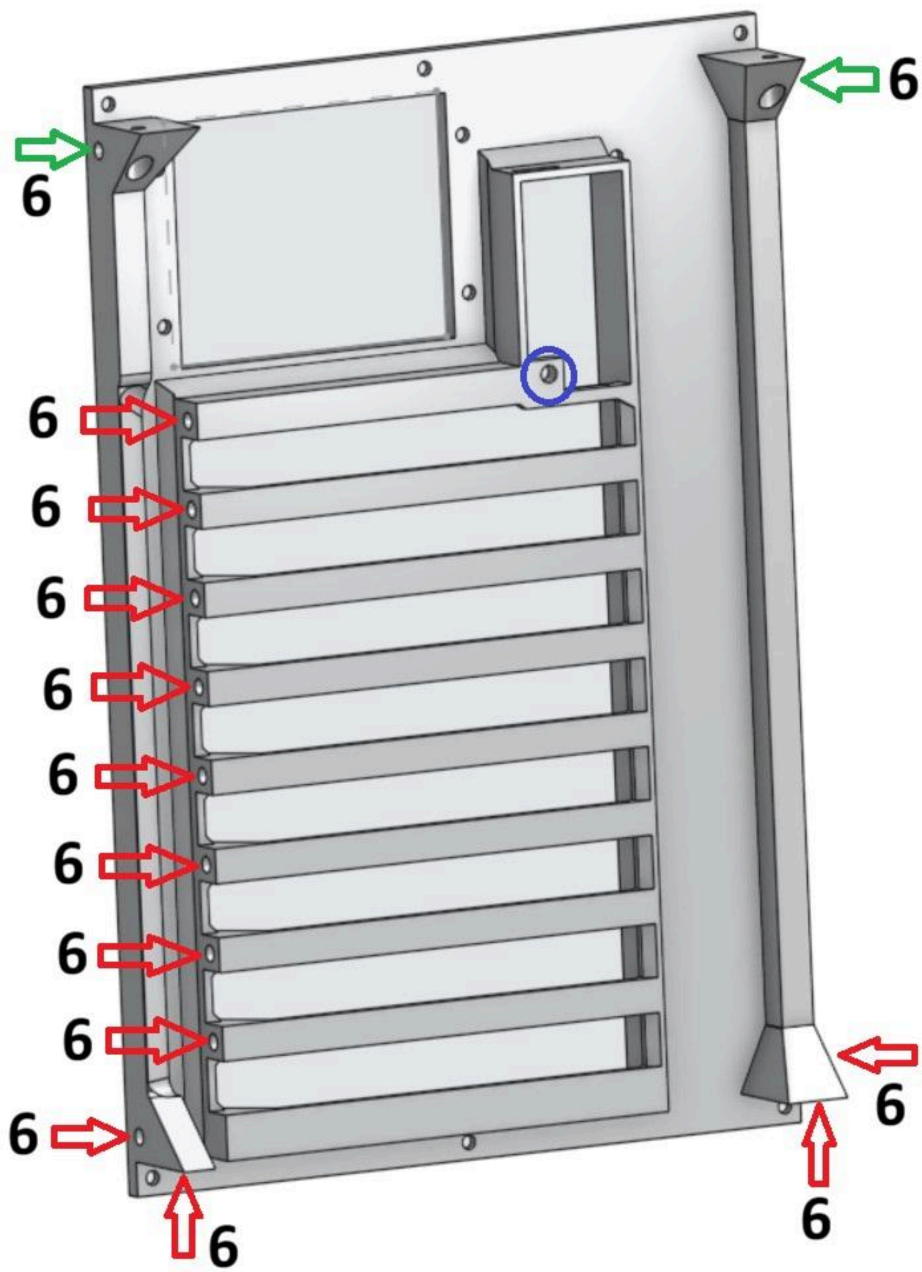
Start with the base frame, by adding the inserts in the spots marked with red.



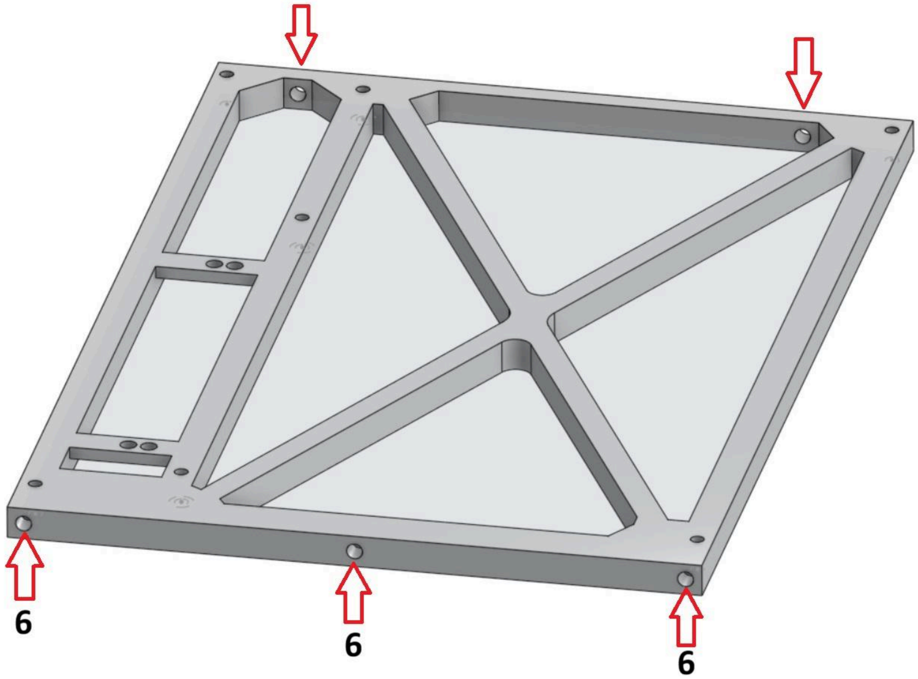
Turn the frame on the backside in order to add the three inserts marked in dark blue. They will be used for securing down the hard drives.

2. Rear frame

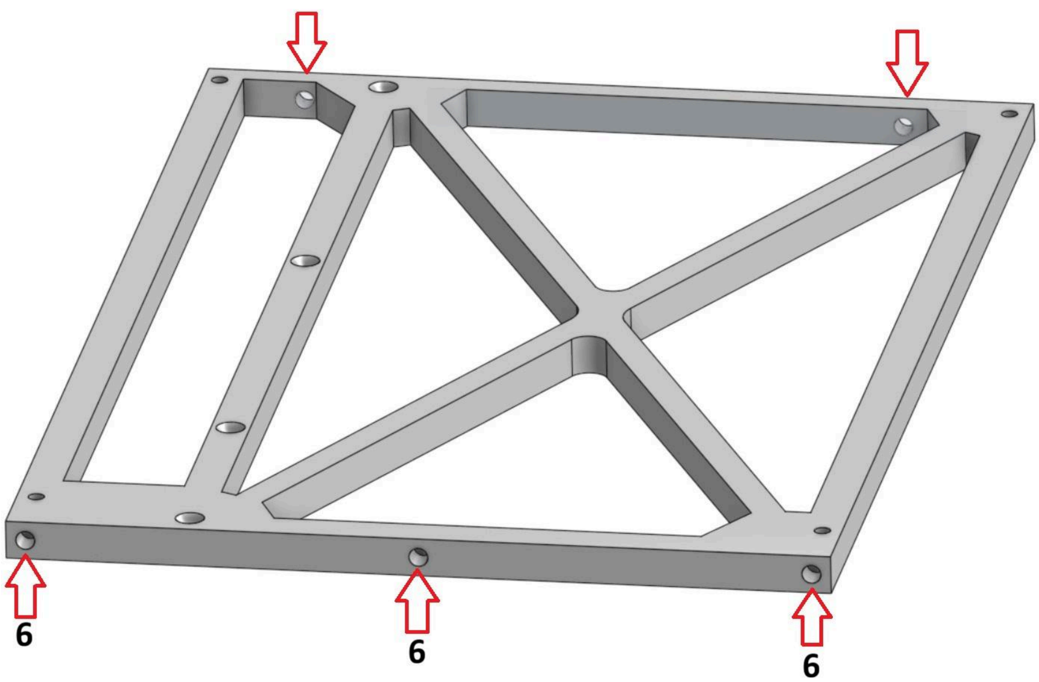
Next, add the inserts on the rear frame as shown. The green spots are optional, if you wish to have additional mounting points for the sidepanels. Flip it on the backside to install the blue spot.



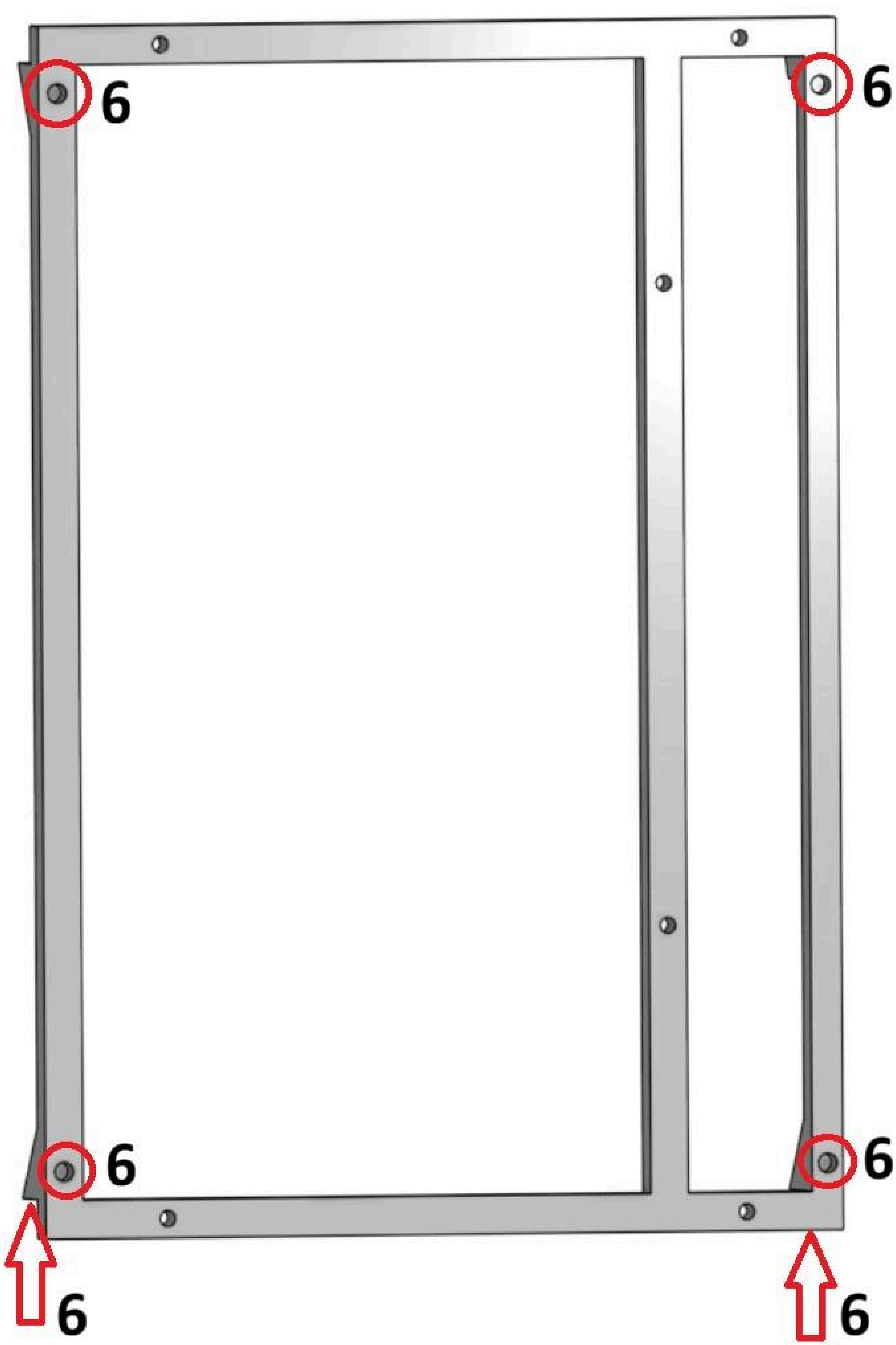
3. Bottom frame



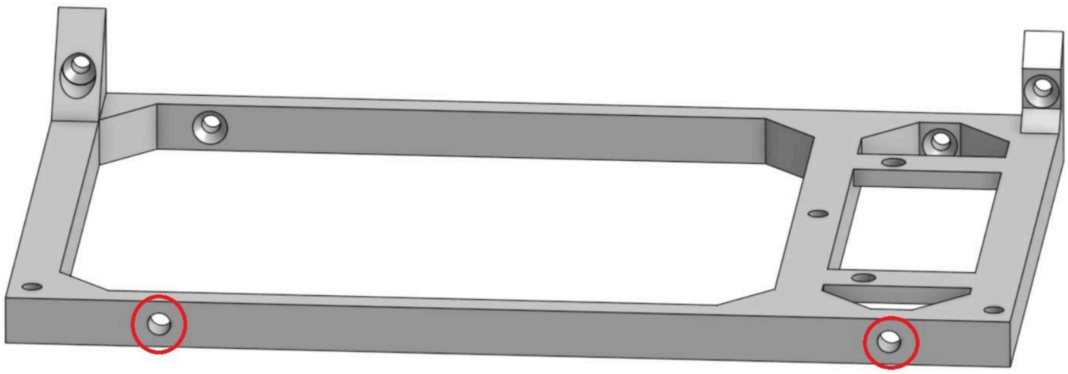
4. Middle frame



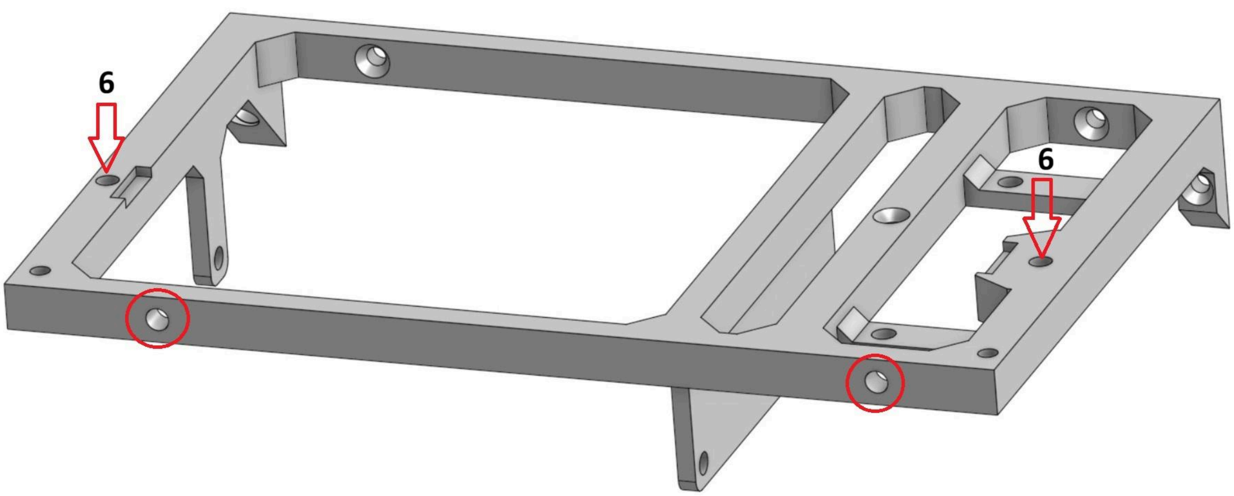
5. Stability frame



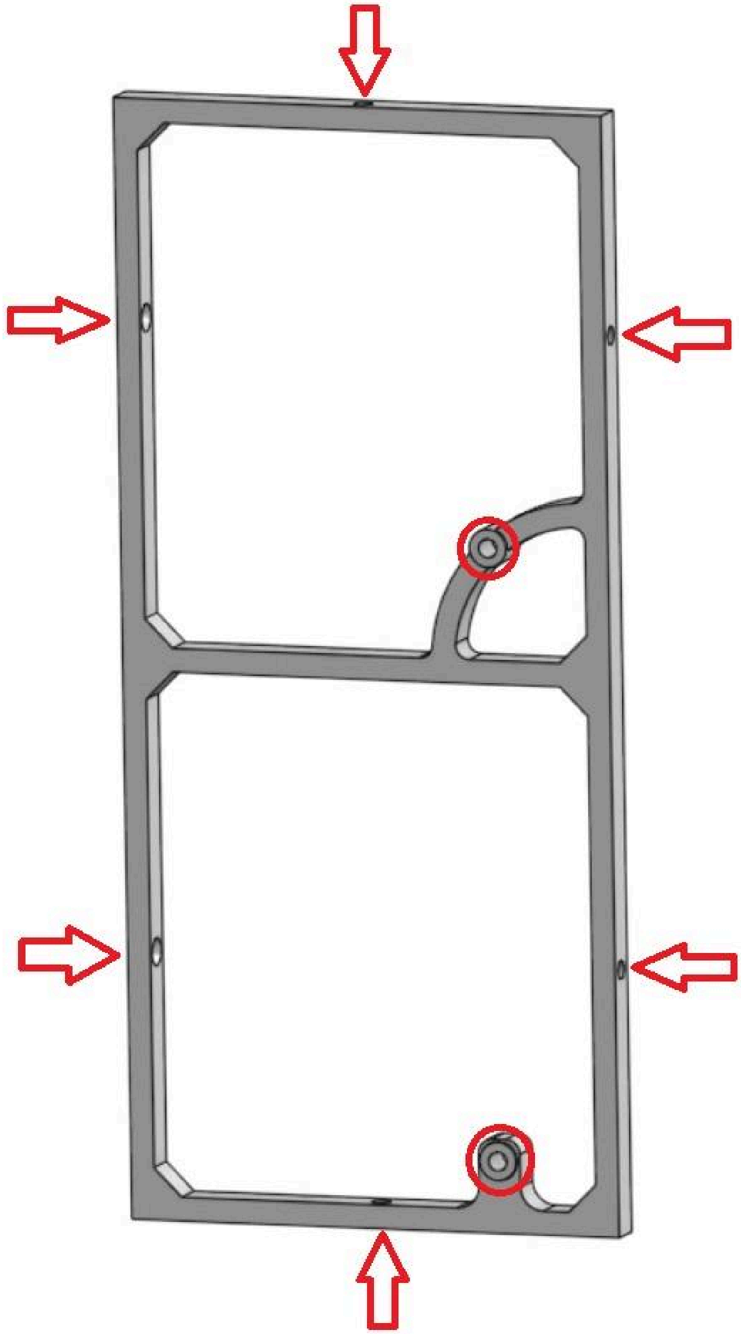
6. Bottom extension frame



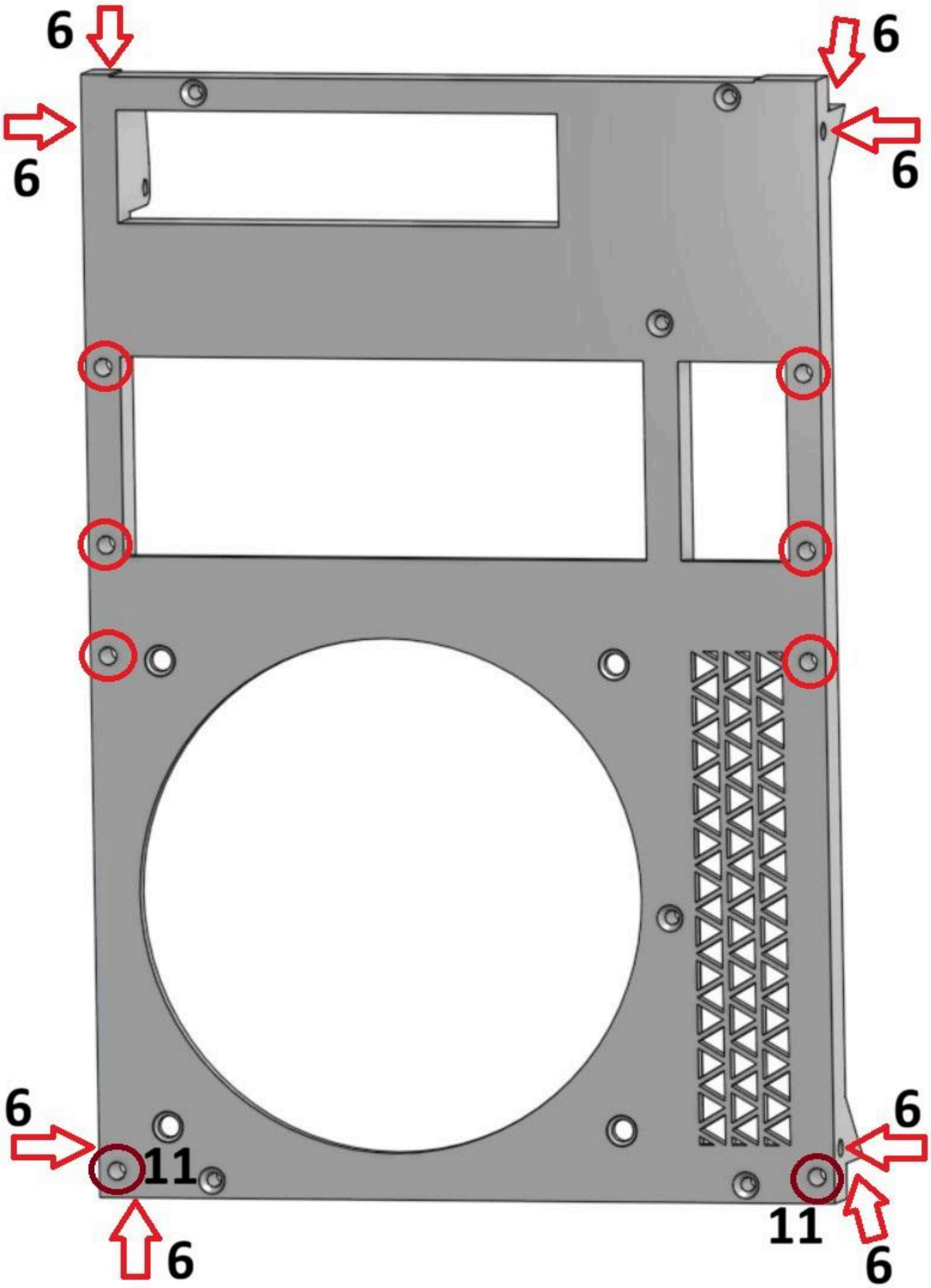
7. Middle extension frame



8. Base extension frame

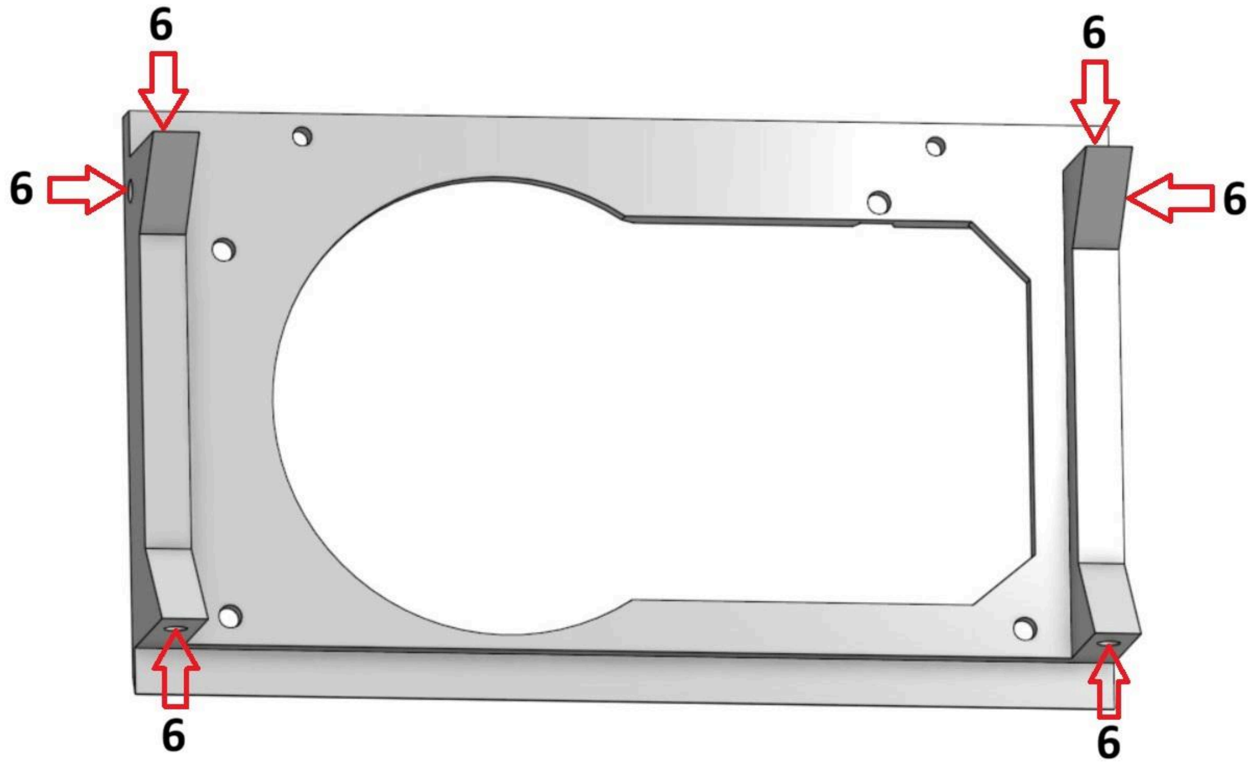


9. Front frame

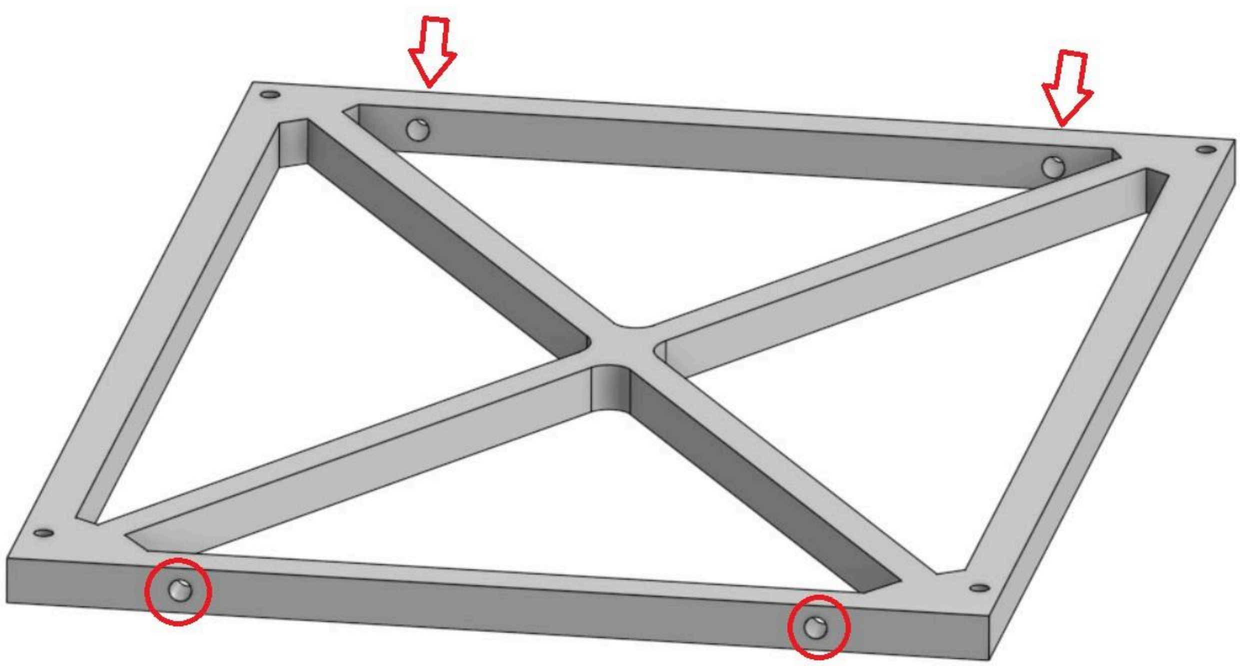


The dark red spots are see-through, but the max insert/screw depth should not exceed 11mm, otherwise it will collide with the rest of the frame.

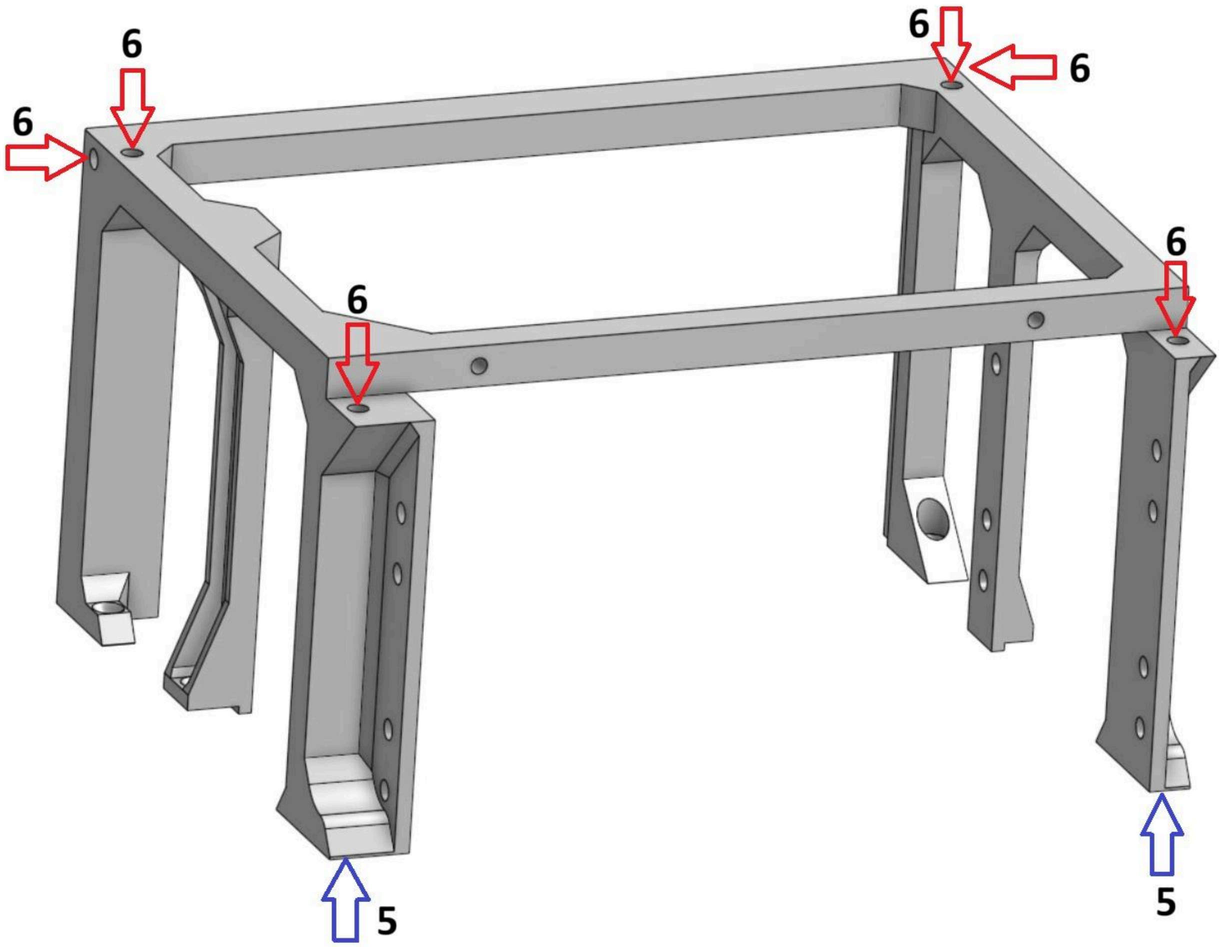
10. PSU frame



11. Top frame



12. Top extension frame



Due to physical constraints, the inserts on the bottom side can only go up to 5mm deep

D. Assembly

All the frame parts have been prepared and are ready to be assembled together.